Limits and Continuity I

1. Evaluate the following. Write "DNE" if the limit does not exist.

$$\lim_{x\to 0}\frac{1}{x^2},\ \lim_{h\to 0}\frac{\sqrt{a+h}-\sqrt{a}}{h},\ \lim_{x\to 2}\frac{x^3-8}{x-2},\ \lim_{x\to 0}\frac{|x|}{x}$$

2. Use the squeeze theorem to evaluate

$$\lim_{x\to 0} x^2\sin\frac{1}{x}.$$

3. **(a)** Show that

$$\lim_{x\to 0^+}\frac{\sin\theta}{\theta}=1$$

using the geometric inequality

$$\cos\theta\le\frac{\sin\theta}{\theta}\le 1,\quad \theta\in(0,\frac{\pi}{2}).$$

(b) Using the fact that $\sin\theta$ is an odd function (i.e., $\sin(-\theta)=-\sin\theta$), conclude that

$$\lim_{x\to 0}\frac{\sin\theta}{\theta}=1.$$

(c) Using the result from part (b), evaluate

$$\lim_{x\to 0}\frac{\sin(ax)}{x}.$$

(d) Again using part (b), evaluate

$$\lim_{x\to 0}\frac{\sin ax}{\sin bx}.$$

4. Suppose two functions $f(x)$ and $g(x)$ are continuous at some point a. **(a)** Show that $(f+g)(x)=f(x)+g(x)$ is continuous at a. **(b)** Show that $(fg)(x)=f(x)\cdot g(x)$ is continuous at a. **(c)** Must f/g be continuous at a? If not, provide a counterexample.

5. Determine all points of discontinuity for each function.

$$f(x)=\frac{x^2-4}{x-2},\quad g(x)=\frac{|x|}{x},\quad h(x)=\begin{cases}\frac{\sin x}{x} & x\ne 0\\ 0 & x=0\end{cases}$$